Downstream Reconstruction and Sparsity in End-to-End Vision Featurizers

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Exploratory block-6 full-grid sweep; one seed; previously analyzed Imagenette test set

Abstract

We implement a registered comparison of local reconstruction, output KL, and output KL plus raw downstream reconstruction for Grassmannian block-sparse featurizers inserted after DINOv3 block 6. All 201 tokens are reconstructed before the remaining frozen transformer blocks and a fixed patch-only Imagenette classifier. Five active-block budgets and three downstream weights define 25 operating points with one shared initialization seed. Allocation failure before main training: 15 of 50 pilot trials completed; 1 additional trial(s) started but did not complete. No main runs or main test evaluations were performed. No LR schedule was frozen. The primary AUC, sparsity thresholds and hypothesis remain unmeasured; this does not falsify or support the directional hypothesis. Recorded stop: The original sweep stage deadline has elapsed. The original deadline was not extended.

1. Context, motivation and related work

Braun et al. [1] optimize output matching and optionally reconstruct downstream layers to distinguish output fidelity from activation fidelity. Their language-model sparsity sweeps motivate this vision adaptation. We retain Goodfire's released tied, signed Grassmannian charts [2,3], rather than conventional ReLU SAEs. The earlier final-layer study [4] is historical evidence, with a different insertion site and token scope. This follow-up does not retrospectively change that experiment.

2. Model, data and objective

The backbone is DINOv3 ViT-B/16 [5], revision 5931719e67bbdb9737e363e781fb0c67687896bc, using its pinned 224-pixel processor. The original model is this backbone plus the previously fitted, frozen ten-class linear readout of maximum absolute final patch coordinates. CLS and four register tokens are excluded from the readout. At the insertion site, however, the BSF replaces every token, preventing a clean-special-token bypass.

We reuse the original duplicate-screened Imagenette partitions: 7,575 training, 1,894 development, and 3,875 test images. The test set was analyzed previously and is explicitly reused; it is not a fresh confirmatory sample. New positional means and one RMS scale are fitted on training-only block-6 states. The complete state is normalized before the BSF and restored to raw units before entering the frozen suffix. No reconstruction error is added back.

Each dictionary contains 4,096 blocks of width 3 (9,441,280 parameters). The retained block counts are k={4,8,16,32,64}, corresponding to {12,24,48,96,192} active coordinates per token. Planned main runs use random seed 200 and identical image order; pilot runs use seed 900. Differentiable QR, tied signed encoder/decoder weights, hard block TopK and one unconstrained learned gain per block follow the released implementation [3].

Local training minimizes normalized insertion MSE. Output training minimizes KL(p_teacher || q_inserted), averaged over images at temperature 1. The downstream conditions use L=KL+(beta/6) sum(j=7..12) MSE(Hhat_j,H_j), where beta={0.0001,0.001,0.01}. Each MSE averages all images, 201 tokens and 768 coordinates in the raw residual state after the indicated block, before terminal normalization. Targets are detached; only BSF parameters train. These conditions contain no insertion-MSE term. Training-variance-normalized downstream errors are diagnostics only.

3. Allocation, optimization and analysis

The new immutable wall-time limit is 24 hours: two for preparation/verification/pilots, training until hour 22, and two for final evaluation/reporting. The registration requires each cell to receive the pilot LR grid {0.0001,0.0003} with seed 900 on fixed class-balanced training/development subsets. Development KL selects LR, with a lower-LR tie break. A complete pilot grid is required before freezing main exposure using measured throughput plus 25% allowance, without consulting test outcomes.

The primary endpoint is local-minus-pure-KL normalized area under KL versus log2 active coordinates, with trapezoidal integration over the five required operating points. Positive differences favor KL. The planned analysis uses 2,000 shared paired image-group bootstrap draws for conditional 95% intervals. There is one training seed, so no seed-level uncertainty or power claim is available. Reconstruction-weight comparisons are descriptive secondary analyses. A mean advantage does not imply dominance at every sparsity or at unmeasured operating points.

4. Results, controls and claims

Execution inventory: 15/50 completed pilots; 16 started pilots; 0/25 started main runs; 0/25 evaluated main runs. The full inventory and individual pilot reports are published with the machine-readable evidence.

Completed pilot development accuracies range from 11.6% to 30.0%. Pure KL has lower development KL in 4/4 completed comparisons matched on k and learning rate. These descriptive tuning-set comparisons use pilot seed 900, are incomplete across sparsity, and provide no main-study or seed-robustness conclusion.

Implementation check on 8 training images: maximum cached/online raw-state difference 0; maximum BSF-gradient difference 1.78814e-07; identity KL 0. Fifty fixed-batch updates reduced recorded KL from 2.29414 to 3.96106e-05. This is an optimization check on training images, not a generalization result.

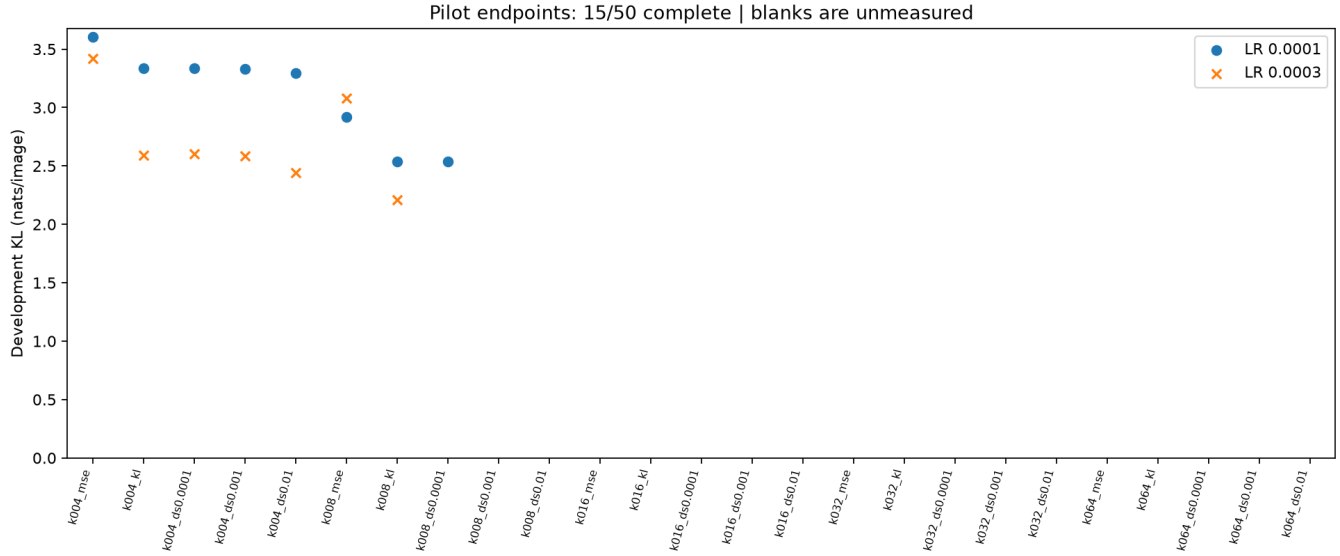
Preparation completed after approximately 68 minutes: extraction took 26.5 minutes, training-only statistics about 23.4 minutes, and final cache hashing about 17.7 minutes. A batch-geometry verification failure stopped the worker at 19:08 UTC and was detected at 19:31 UTC. The repaired check passed at 19:37 UTC; pilots resumed at 19:38 UTC. The two-hour preparation deadline stopped the 16th trial during development evaluation at 20:00:02 UTC. Fifteen trials completed, one had trained but lacked a development endpoint, and 34 never started. No main schedule, selected learning rates, main training or test evaluation followed. The bottleneck was the preparation allocation, not a measured scientific failure of KL training.

No learning rate was selected for main training, and no main-study bootstrap interval, threshold, Pareto frontier, test control, map or intervention was calculated. These quantities are unmeasured. The allocation failure is an engineering outcome and neither supports nor falsifies the primary scientific hypothesis.

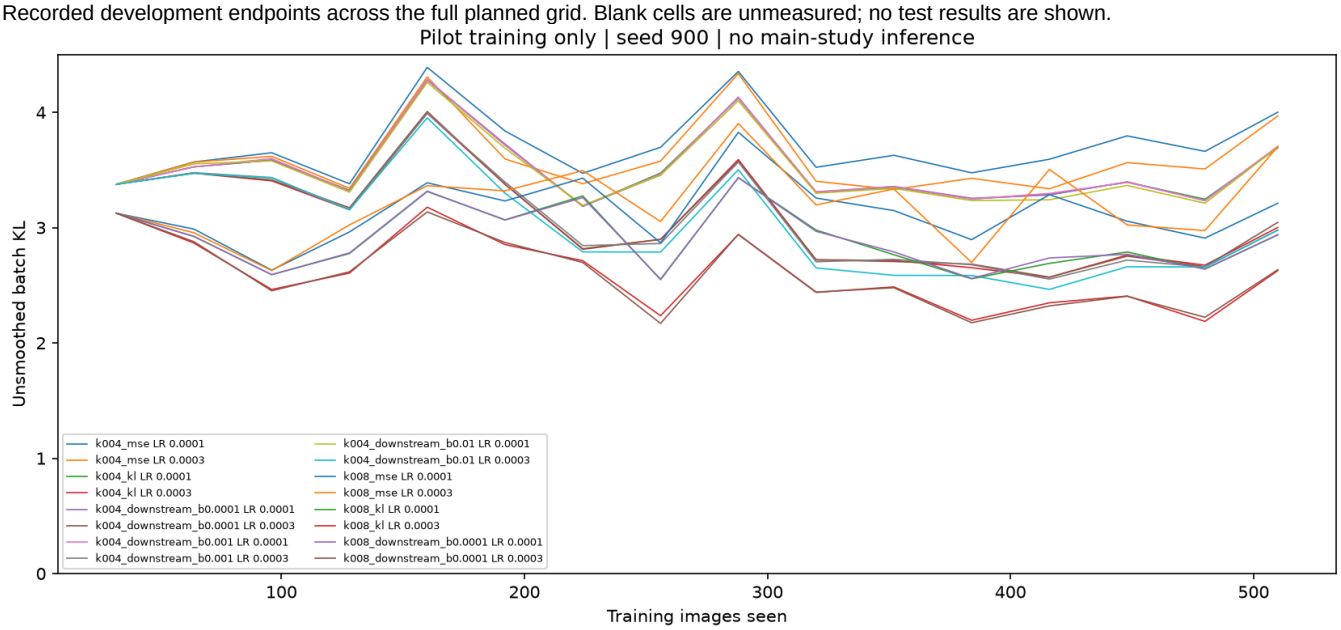
Pilot objective	k	LR	Dev KL	Dev accuracy
Local MSE	4	0.0001	3.6066	12.40%
Local MSE	4	0.0003	3.42204	12.00%
Output KL	4	0.0001	3.3338	14.00%
Output KL	4	0.0003	2.58942	21.60%
KL + 0.0001 downstream MSE	4	0.0001	3.33407	13.60%
KL + 0.0001 downstream MSE	4	0.0003	2.60219	22.40%
KL + 0.001 downstream MSE	4	0.0001	3.3304	13.60%
KL + 0.001 downstream MSE	4	0.0003	2.58754	22.00%
KL + 0.01 downstream MSE	4	0.0001	3.29245	15.20%
KL + 0.01 downstream MSE	4	0.0003	2.44361	22.80%
Local MSE	8	0.0001	2.91783	13.20%

Pilot objective	k	LR	Dev KL	Dev accuracy
Local MSE	8	0.0003	3.08239	11.60%
Output KL	8	0.0001	2.53897	20.00%
Output KL	8	0.0003	2.21152	30.00%
KL + 0.0001 downstream MSE	8	0.0001	2.53986	22.00%

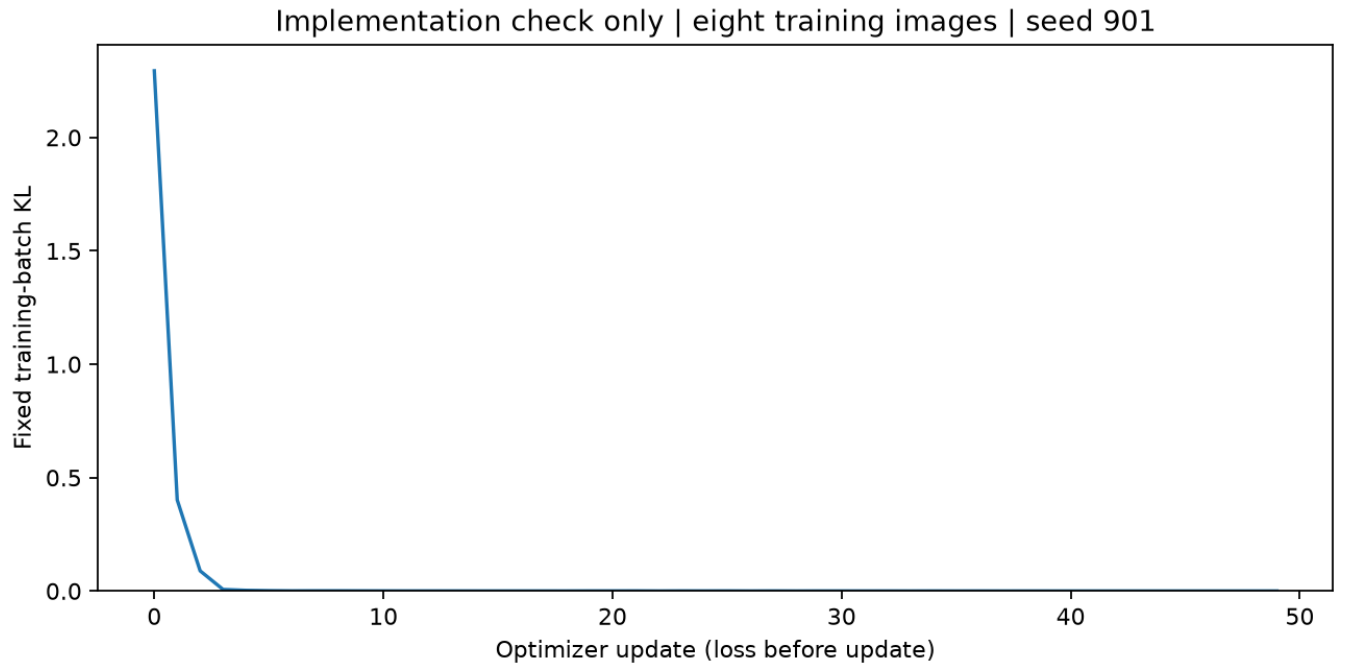
Table: all completed pilot endpoints on 250 development images after 510 training images. The incomplete grid and tuning-set measurements do not establish objective superiority.



Recorded development endpoints across the full planned grid. Blank cells are unmeasured; no test results are shown.



Every started pilot's recorded batch KL. Learning rates and seed belong to pilots only.



Fixed training-batch optimization check, separate from all pilot and main conditions.

5. Interventions, discussion and limitations

The registered k=16 diagnostic protocol selects blocks by development-patch energy for fixed development examples. Removal and predetermined same-class/other-class donor swaps alter the selected block over all 201 positions. The random control samples among eight nearest-energy active alternatives. Matching is approximate, and inactive selected blocks must remain in results. This protocol defines interventions on the composite classifier, not evidence of semantic correspondence. It was not executed because main training did not start.

A ten-class readout constrains only a narrow portion of DINO behavior. Lower KL, lower reconstruction error, or sparse codes do not establish full representation preservation, genuine concepts, or faster inference. Raw downstream MSE can emphasize high-variance layers and special tokens; separate patch/special and normalized diagnostics expose this issue. One seed, short training, reused test data, and a two-value LR search limit generalization. The registered main comparison would include development-selected LRs; none were selected in this allocation. Known study duplicates are isolated; unknown backbone pretraining overlap is unresolved.

6. Conclusion

Allocation failure before main training: 15 of 50 pilot trials completed; 1 additional trial(s) started but did not complete. No main runs or main test evaluations were performed. No LR schedule was frozen. The primary AUC, sparsity thresholds and hypothesis remain unmeasured; this does not falsify or support the directional hypothesis. Recorded stop: The original sweep stage deadline has elapsed. The original deadline was not extended.

References and reproducibility

1. Braun et al. End-to-End Sparse Dictionary Learning (2024), Sections 2.4 and 3

2. Fel et al. Block-Sparse Featurizers (2026)

3. Goodfire implementation, revision 219f121

4. Project, original study, full artifacts and preregistration

5. DINOv3 implementation

6. Imagenette dataset

The publication format includes BSF weights, the fixed classifier, normalization, predictions, histories and provenance. Original photos, raw activation caches and gated backbone weights are excluded. Local checkpoints preserve optimizer and RNG state. No completed main-study bundle is eligible for publication in this failed allocation.

AI statement

Codex wrote this codebase and paper. Human cleanup and rewriting are in progress. A human provided the exact specification that Codex executed.